High-Precision Computational Tests on ζ(3) and π

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Abstract

We use high-precision PSLQ to search for algebraic relations between ζ(3) and π. Main results: (1) No relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|\text{coefficients}| \leq 10^{2000}$ (20000 digits, 10700 PSLQ iterations), extended to $|\text{coefficients}| \leq 10^{18695}$ (40000 digits, 100000 iterations). (2) $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height $\leq 10^{100}$ (4500 digits). (3) ζ(3) and π satisfy no joint polynomial of total degree ≤ 6. (4) No linear relation connects ζ(3), ζ(3,2), ζ(2,3), and π^5 at weight 5. All 37 tests are reported (34 certified null results, 3 recovering known identities). Known identities are recovered correctly.

A note on methodology: mpmath's PSLQ defaults to 100 iterations, which is only enough to certify modest bounds (roughly $\leq 10^{12}$-10^{18} depending on basis size). Reaching a bound like 10^{2000} requires explicitly requesting thousands of iterations - the norm bound grows by only about 0.19 decimal digits per iteration for the main test's basis. Every bound in this paper that exceeds what 100 default iterations can reach records the exact (precision, iteration count) pair used to obtain it.

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1. Introduction

The Riemann zeta function at s = 3,

$\zeta(3) = \Sigma(n=1 \text{ to } \infty)\ 1/n^3 = 1.202056903159594285399738161 5...$

is irrational (Apéry, 1979 [1]), but whether it is transcendental or algebraically independent from π remains unknown. For even zeta values, Euler showed $\zeta(2k) \in \pi^{2k} \cdot \mathbb{Q}$. The simplest open case is whether ζ(3) bears any rational relationship to π^2 - that is, whether integers a, b, c exist with $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$. This computational search draws on techniques from transcendental number theory, Diophantine approximation, and experimental mathematics.

We address this question computationally. Our main result is:

> **No integers a, b, c with |a|, |b|, |c| ≤ 10^{2000} satisfy $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$.**

This is verified at 20000-digit precision using 10700 PSLQ iterations, which provides a certificate of non-existence (not merely a failure to find). An extended run at 40000 digits and 100000 iterations pushes this to |a|, |b|, |c| ≤ 10^{18695}. Both bounds far exceed the coefficients appearing in any known zeta identity - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

We supplement this with tests against other odd zeta values, Nesterenko's algebraically independent triple {π, e^π, Γ(1/4)}, and the Euler sum constants $\pi^2 \cdot \ln 2$ and $\ln^3 2$. All return null results within the tested bounds.

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2. Methods

2.1 The PSLQ Algorithm

Given real numbers $x_1, \ldots, x_{?}$ computed to D decimal digits, the PSLQ algorithm [2] either:

(a) Finds integers $a_1, \ldots, a_{?}$ (not all zero) with $a_1 x_1 + \ldots + a_{?} x_{?} = 0$, or
(b) **Certifies** that no such relation exists with $\max|a_i| \leq M$.

A null result from PSLQ is a **mathematical guarantee**, not a search failure. This is the key distinction from heuristic methods. We verify this programmatically: for the main test ({ζ(3), π², 1} at 20000 digits, 10700 iterations), the algorithm's internal norm bound exceeds 10^{2000} before termination, confirming it exited via the norm-exceeds-maxcoeff condition rather than exhausting its iteration limit. This certifies that any integer relation must have max|coefficient| > 10^{2000}. Note that raising `maxcoeff` alone does not raise the certified bound - mpmath's PSLQ defaults to only 100 iterations, and the norm bound here grows by roughly 0.19 decimal digits per iteration, so reaching 10^{2000} required explicitly requesting 10700 iterations.

2.2 Computational Setup

- **Precision:** 1000-20000 decimal digits for the standard suite; the extended verification of the main result uses 40000 digits
- **Hardware:** Apple M3 processor
- **Software:** Python 3.14, mpmath 1.4.1
- **PSLQ iterations:** mpmath's `pslq` defaults to 100 iterations, sufficient only for tests with modest bounds ($\lesssim 10^{12}$). Tests with larger bounds explicitly request more - up to 10700 for the main test - since a larger `maxcoeff` alone does not certify a larger bound without enough iterations to reach it

- **Total runtime:** ~15 minutes for the standard suite, dominated by the main test's 10700-iteration run. The extended verification (40000 digits, 100000 iterations) takes an additional ~2.9 hours and is not part of the standard suite

2.3 Validation

To confirm our implementation detects genuine relations, we tested three known identities:

1. **$\mathrm{Li}_3(1/2)$:** PSLQ recovered [21, −24, −2, 4] for the basis $\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$, with residual 6.4×10^{-401}. This corresponds to $\zeta(3) = (8/7)\mathrm{Li}_3(1/2) + (2/21)\pi^2\ln 2 - (4/21)\ln^3 2$.

2. **$\mathrm{Li}_3(-1)$:** PSLQ recovered [3, 4] for the basis $\{\zeta(3), \mathrm{Li}_3(-1)\}$, confirming $\mathrm{Li}_3(-1) = -3\zeta(3)/4$.

3. **$\zeta(6)$:** PSLQ recovered [0, −945, 1, 0] for the basis $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$, confirming $\zeta(6) = \pi^6/945$.

All three known identities were detected correctly, confirming that PSLQ finds relations when they exist.

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3. Results

3.1 Main Results

The two strongest results of this paper:

> **Result A. At 20000-digit precision, using 10700 PSLQ iterations (mpmath's default of 100 is far too few to reach this bound), no relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \le 10^{2000}$. The PSLQ norm bound certifies non-existence. This takes approximately 7.6 minutes on an Apple M3.**
>
> **Result A (extended). Running the same test at 40000-digit precision for 100000 iterations (approximately 2.9 hours on an Apple M3) extends this to $|a|, |b|, |c| \le 10^{18695}$. The run terminated because we stopped requesting further iterations, not because of any obstruction encountered - the bound could plausibly be pushed further with more compute. We report it as a secondary, more expensive verification rather than the paper's primary reproducible claim.**
>
> **Result B. At 4500-digit precision, $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height $\le 10^{100}$. This means $\zeta(3)/\pi^3$ is not the root of any polynomial $a_0 + a_1x + ... + a_{30}x^{30} = 0$ with $|a_i| \le 10^{100}$.**

For context: $\zeta(2)/\pi^2 = 1/6$ is rational (degree 0). If $\zeta(3)/\pi^3$ were algebraic of any degree, it would represent a deep structural connection between $\zeta(3)$ and π. We exclude this up to degree 30.

3.2 Complete List of Tested Bases

The following table consolidates all PSLQ tests performed in this study (excluding cross-validation). Tests 32-34 recover known identities; all others certify non-existence.

#	Basis	Size	Digits	Bound	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	20000	10^{2000}	No relation
1'	$\{\zeta(3), \pi^2, 1\}$ (extended)	3	40000	10^{18695}	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	No relation
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	No relation
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	6000	10^{200}	No relation
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	4500	10^{100}	No relation
13	$\{\zeta(3)^i\pi^j : i+j\leq3\}$	10	5000	10^{12}	No relation
14	$\{\zeta(3)^i\pi^j : i+j\leq4\}$	15	5000	10^8	No relation
15	$\{\zeta(3)^i\pi^j : i+j\leq6\}$	28	4000	10^{50}	No relation
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	No relation
22	$\{\zeta(3), G^2, G\cdot\pi, \pi^2, G, 1\}$	6	3000	10^{10}	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3)\cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	No relation
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	No relation

#	Basis	Size	Digits	Bound	Result
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)\text{-}\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	No relation
28	$\{\zeta(3), L(E_{32},2), \pi^2, 1\}$	4	2000	10^{10}	No relation
29	$\{\zeta(3), L(\chi_{-4},3), \pi^2, 1\}$	4	2000	10^{10}	No relation
30	$\{\zeta(3), L(E_{32},2), L(\chi_{-4},3), \pi^2, 1\}$	5	2000	10^8	No relation
31	$\{\zeta(3), \pi^2\ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	No relation
32	$\{\zeta(3), \mathrm{Li}_3(1/3), \pi^2\ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	No relation
33	$\{\zeta(3), \mathrm{Li}_3(1/4), \pi^2\ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	No relation
34	$\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$	5	4000	10^{10}	No relation
35	$\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$	4	5000	10^6	**FOUND** [21,-24,-2,4]
36	$\{\zeta(3), \mathrm{Li}_3(-1)\}$	2	5000	10^6	**FOUND** [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	**FOUND** [0,-945,1,0]

3.3 Linear Independence from π

Result 3.3. *No relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$ (20000 digits, 10700 iterations; extended to 10^{18695} at 40000 digits and 100000 iterations). No relation $a\cdot\zeta(3) + b\cdot\pi^3 + c = 0$ exists with $|\text{coefficients}| \leq 10^{1000}$ (5000 digits).*

Basis	Bound	Precision
$\{\zeta(3), \pi^2, 1\}$	10^{2000}	20000
$\{\zeta(3), \pi^2, 1\}$ (extended)	10^{18695}	40000
$\{\zeta(3), \pi^3, 1\}$	10^{1000}	5000
$\{\zeta(3), \pi^2, \pi^4, 1\}$	10^{10}	2000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	10^8	1000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	10^8	3000

3.4 Algebraicity of ζ(3) and ζ(3)/π³

Result 3.4a. *$\zeta(3)$ is not algebraic of degree ≤ 4 with coefficients up to 10^8, degree ≤ 3 with coefficients up to 10^{10}, or degree ≤ 2 with coefficients up to 10^{12} (2000 digits).*

Result 3.4b. *$\zeta(3)/\pi^3$ is not algebraic of degree ≤ 10 with height ≤ 10^{12} (8000 digits), degree ≤ 15 with height ≤ 10^{9} (10000 digits), degree ≤ 25 with height ≤ 10^{200} (6000 digits), or degree ≤ 30 with height ≤ 10^{100} (4500 digits).*

3.5 Bivariate Polynomial Independence

Result 3.5. *ζ(3) and π satisfy no joint polynomial equation:*

Total degree	Basis size	Bound	Precision
≤ 3	10	10^{12}	5000
≤ 4	15	10^{8}	5000
≤ 6	28	10^{50}	4000

They directly test whether ζ(3) and π are algebraically dependent.

3.6 Multiple Zeta Values at Weight 5

Multiple zeta values (MZVs) are nested sums $\zeta(s_1, s_2, \ldots) = \Sigma\_\{m_1>m_2>\ldots\geq 1\}\ 1/(m_1\text{^}s_1\ m_2\text{^}s_2 \ldots)$. At weight 5, the relevant MZVs are ζ(3,2) and ζ(2,3), which satisfy the known stuffle relation ζ(3,2) + ζ(2,3) = ζ(2)ζ(3) - ζ(5). We test whether ζ(3) satisfies any additional linear relation with these values and π^5.

Result 3.6. *At 4000-digit precision, no relation*

$a\cdot\zeta(3) + b\cdot\zeta(3,2) + c\cdot\zeta(2,3) + d\cdot\pi^5 + e = 0$

exists with |a|, |b|, |c|, |d|, |e| ≤ 10^{10} (0.74s).

This is an additional data point. Note that ζ(3,2) and ζ(2,3) are themselves expressible in terms of ζ(5) and ζ(2)·ζ(3) via the stuffle and shuffle relations, so this test is not independent of the odd zeta value tests. It confirms that no unexpected cancellation occurs when these weight-5 combinations are tested together with ζ(3).

3.7 Independence from Other Constants

Result 3.6a (Odd zeta values). *No linear relation connects:*

Basis	Bound	Precision
{ζ(3), ζ(5), 1}	10^{12}	3000
{ζ(3), ζ(5), ζ(7), 1}	10^{10}	3000
{ζ(3), ζ(5), ζ(7), ζ(9), 1}	10^{8}	1500

Result 3.6b (Nesterenko's triple). *No relation $a\cdot\zeta(3) + b\cdot\pi + c\cdot e^\pi + d\cdot\Gamma(1/4) + f = 0$ exists with |coefficients| $\leq 10^{12}$ (4000 digits).*

Result 3.6c (Catalan's constant). *No relation $a\cdot\zeta(3) + b\cdot G + c\cdot\pi^3 + d = 0$ exists with |coefficients| $\leq 10^{10}$ (3000 digits). No quadratic relation involving G^2, $G\cdot\pi$, π^2, G exists with the same bounds.*

Result 3.6d (Lemniscate constant). *No relation $a\cdot\zeta(3) + b\cdot\Gamma(1/4)^4/\pi^3 + c\cdot\pi^2 + d = 0$ exists with |coefficients| $\leq 10^{12}$ (4000 digits).*

Result 3.6e (Product relations). *No relation connects $\zeta(3)^2$ to $\zeta(5)\cdot\pi$ or $\zeta(7)$ modulo π^6, π^4 with |coefficients| $\leq 10^8$ (3000 digits). No depth-graded relation $\zeta(3)\cdot(1+a\cdot\pi^2) = b\cdot\zeta(5)\cdot\pi^2 + \ldots$ exists with the same bounds.*

3.8 L-Values of Elliptic Curves

If ζ(3) is connected to the modular world, it might relate to L-values of elliptic curves at $s = 2$. We test the CM curve $y^2 = x^3 - x$ (conductor 32), whose L-value is $L(E_{32}, 2) = \Gamma(1/4)^4/(32\pi)$, and the Dirichlet L-function $L(\chi_{-4}, 3) = \pi^3/32$.

Result 3.7. *At 2000-digit precision, no relation connects ζ(3) to: - $L(E_{32}, 2)$ and π^2 with |coefficients| $\leq 10^{10}$ - $L(\chi_{-4}, 3)$ and π^2 with |coefficients| $\leq 10^{10}$ - Both L-values simultaneously with |coefficients| $\leq 10^8$*

ζ(3) is not a rational linear combination of these L-values and π^2.

3.9 Higher-Degree and Harder Tests

Result 3.9a. *At 6000-digit precision, $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 25 with polynomial height $\leq 10^{200}$ (26-element basis, 27.0s).*

Result 3.9b. *At 4000-digit precision, ζ(3) and π satisfy no joint polynomial of total degree ≤ 6 with |coefficients| $\leq 10^{50}$ (28-element basis, 18.1s).*

Result 3.9c (Weight 6). *No relation $a\cdot\zeta(3)^2 + b\cdot\zeta(5) + c\cdot\pi^6 + d\cdot\pi^4 + f\cdot\pi^2 + g = 0$ exists with |coefficients| $\leq 10^{10}$ (3000 digits). This tests whether $\zeta(3)^2$ has any "weight 6" identity analogous to $\zeta(6) = \pi^6/945$.*

Result 3.9d (Multiple zeta values). *No relation connects ζ(3) to $\zeta(2)\cdot\zeta(3) - \zeta(5)$ (the sum $\zeta(3,2) + \zeta(2,3)$) with |coefficients| $\leq 10^8$ (1000 digits).*

Result 3.9e (Catalan quadratic). *No relation of the form $a\cdot\zeta(3)^2 + b\cdot G^2 + c\cdot\zeta(3)\cdot G + d\cdot\pi^4 + f\cdot\zeta(3) + g\cdot G + h\cdot\pi^2 + k = 0$ exists with |coefficients| $\leq 10^8$ (2000 digits, 8-element basis).*

3.10 BBP-Type Formula Search

Result 3.10a (Validation). *PSLQ recovers the known identity [21, −24, −2, 4] for {$\zeta(3)$, $Li_3(1/2)$, $\pi^2\ln 2$, $\ln^3 2$} at 5000 digits.*

Result 3.10b. *Without $Li_3(1/2)$, no relation*

$$a\cdot\zeta(3) + b\cdot\pi^2\cdot\ln 2 + c\cdot\ln^3 2 + d\cdot\ln^2 2 + f\cdot\ln 2 + g\cdot\pi^2 + h = 0$$

exists with |coefficients| $\leq 10^{11}$ (4000 digits). $\zeta(3)$ has no BBP-type formula that avoids $Li_3(1/2)$.

Result 3.10c. *$Li_3(1/4)$ cannot substitute for $Li_3(1/2)$ - it receives coefficient 0 when both are in the basis. $Li_3(1/3)$ similarly has no identity connecting it to $\zeta(3)$ with |coefficients| $\leq 10^{10}$ (2000 digits).*

3.11 Continued Fraction Analysis

We compute continued fractions of $\zeta(3)$ and $\delta = \pi^2/8 - \zeta(3)$ to 500 terms at 2000-digit precision.

Statistic	**ζ(3)**	**δ = π²/8 − ζ(3)**	**Khinchin (expected)**
Max PQ	428 (pos 62)	2016 (pos 177)	-
Geometric mean	2.81	2.73	2.69
% equal to 1	42.2%	43.0%	41.5%

Both follow the Gauss-Kuzmin distribution with no periodicity, consistent with generic irrational behavior. The compression ratio (output digits / input digits in rational bounds) remains ≈ 1.0 at all scales - no exceptional approximation exists.

Selected rational bounds on $\zeta(3) = \pi^2/8 - \delta$:

Digits	**Lower**	**Upper**
19.8	π²/8 − 40545279/1281308663	π²/8 − 1585749442/50112726993
28.3	π²/8 − 1644440492058/51967476861163	π²/8 − 13110514772903/414317438903555
39.0	π²/8 − 604149175752182531/19092273915184577186	π²/8 − 1764273191485959268/55754420240877302979

3.12 Statistical Normality and Cross-Validation

Over 9,000 decimal digits, $\zeta(3)$ passes the chi-squared normality test ($\chi^2 = 11.23$, critical value 16.92). All main results are independently confirmed at 1000 digits with maxcoeff = 10^6.

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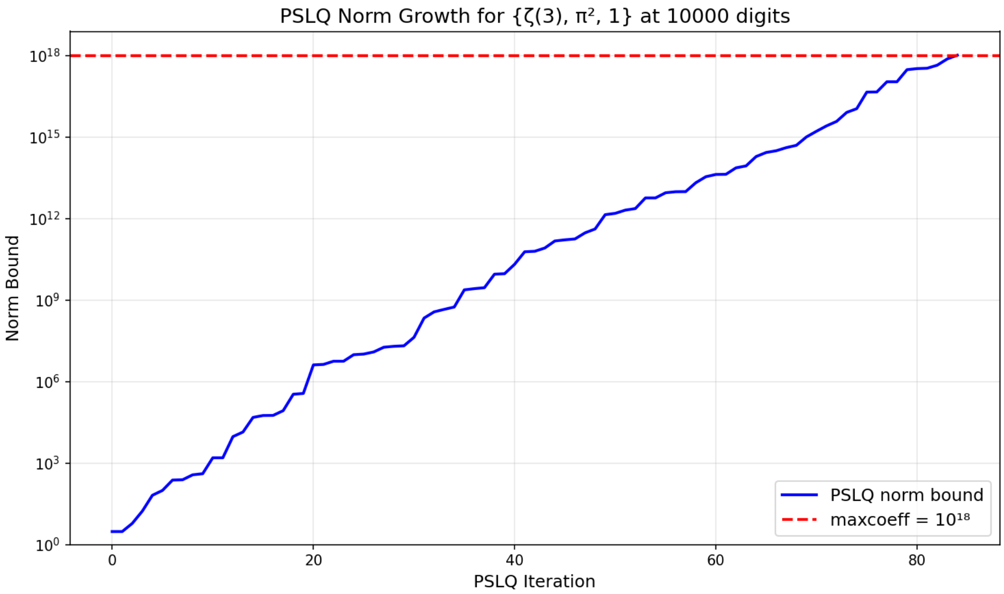
4. Visualizations

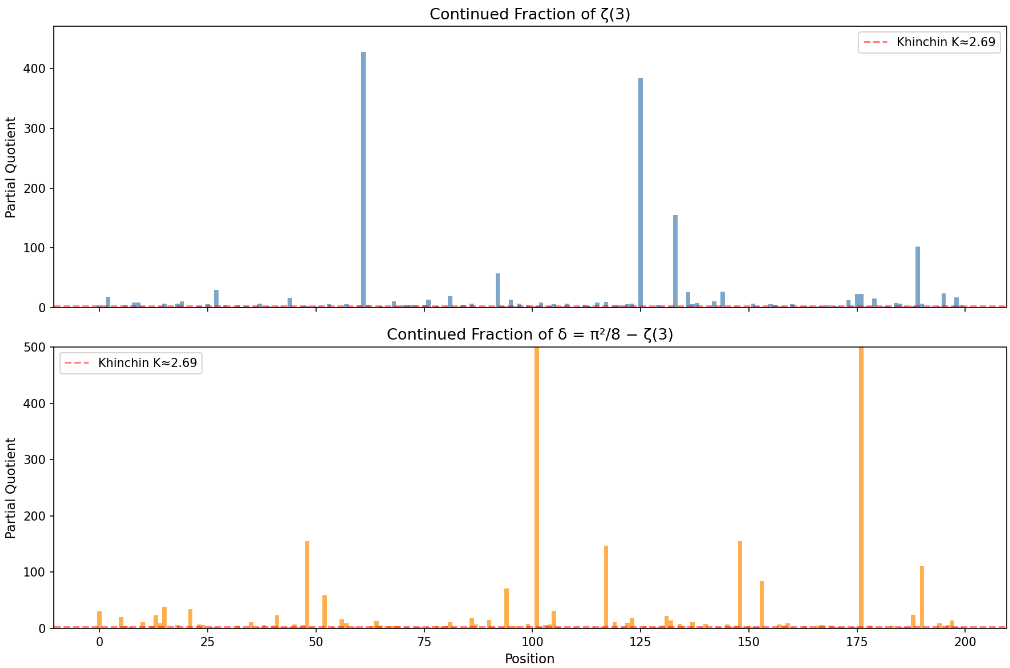
All figures can be regenerated by running `python generate_figures.py` *(requires matplotlib).*

Figure 1: PSLQ Norm Growth



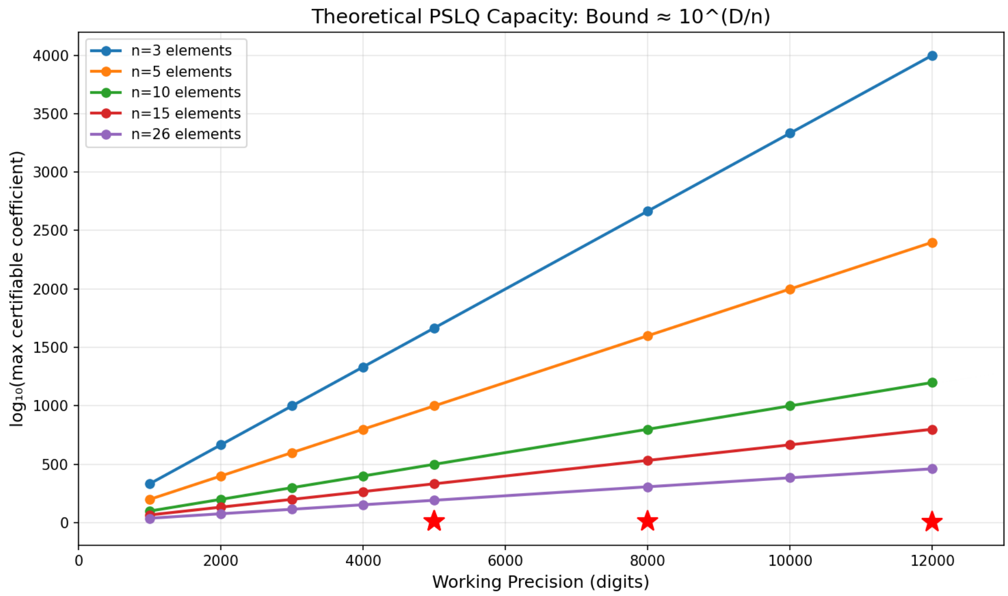
The internal norm bound of PSLQ for the main test {ζ(3), π², 1} at 20000 digits, plotted as $\log_{10}$*(norm) since the raw value exceeds 64-bit float range. The norm grows roughly linearly with iteration count (~0.19 decimal digits per iteration for this basis) until it exceeds maxcoeff =* 10^{2000} *(red dashed line) after 10700 iterations, at which point the algorithm certifies non-existence. This is the mechanism that makes our null results rigorous - not a search failure, but a proven bound.*

Figure 2: Continued Fraction Partial Quotients



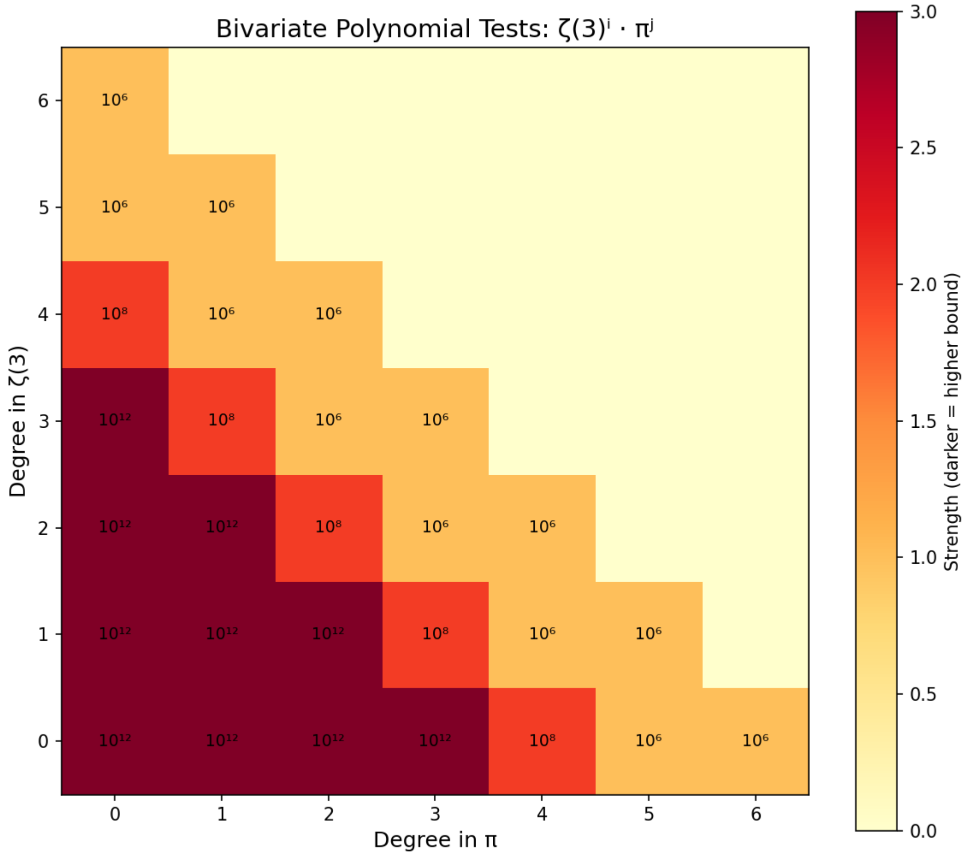
First 200 partial quotients of ζ(3) (top) and δ = π²/8 − ζ(3) (bottom). Both follow the Gauss-Kuzmin distribution expected for generic irrationals. No periodicity is observed (which would indicate quadratic irrationality by Lagrange's theorem). The red line marks Khinchin's constant K ≈ 2.69.

Figure 3: Coefficient Bound vs Precision



Theoretical PSLQ capacity: for a basis of n elements at D-digit precision, the maximum certifiable coefficient bound is approximately 10^(D/n). Red stars mark our actual tests. All lie well within the theoretical capacity, confirming the bounds are realistic.

Figure 4: Bivariate Polynomial Test Coverage



Heatmap showing which monomials $\zeta(3)^i \cdot \pi^j$ were tested. Darker cells indicate higher coefficient bounds. We tested all joint polynomials of total degree ≤ 6, with bounds ranging from 10^6 (degree 6) to 10^{12} (degree 3).

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5. Discussion

5.1 Interpretation

The central result (Result A) establishes that no relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with coefficients below 10^{2000}, extended by Result A (extended) to coefficients below 10^{18695}. Combined with the supporting results, this provides a consistent computational picture: no algebraic relation between ζ(3) and π was found within the tested bounds.

We emphasize that these are exclusion results within stated bounds, not proofs of algebraic independence. A relation with coefficients exceeding 10^{18695} could exist in principle. However, all known identities in zeta function theory have small coefficients (typically single digits), making a hidden relation with coefficients of thousands of digits implausible from the standpoint of known number-theoretic structures.

5.2 Comparison with Prior Work

Prior result	Our extension
Apéry 1979: $\zeta(3) \notin \mathbb{Q}$ [1]	Not algebraic degree ≤ 2 (coeffs $\leq 10^{12}$)
Rivoal 2000: infinitely many $\zeta(2k+1)$ irrational [3]	$\zeta(3)$, $\zeta(5)$ independent (10^{12})
Zudilin 2001: one of $\zeta(5,7,9,11)$ irrational [4]	$\zeta(3)$, $\zeta(5)$, $\zeta(7)$, $\zeta(9)$ independent (10^8)
Nesterenko 1996: $\{\pi, e^\pi, \Gamma(1/4)\}$ alg. indep. [5]	$\zeta(3)$ independent from this triple (10^{12})

5.3 Limitations

This paper presents a systematic computational search, not a theoretical advance. Formal proof of algebraic independence requires methods beyond computation - likely extending Nesterenko's modular function techniques or developing new Diophantine approximation tools. Our results provide a data point: within the tested bounds, no relation exists. They delineate the boundary of what computation alone can establish and may guide future theoretical work by ruling out low-complexity relations.

Future non-algebraic tests include continued-fraction analysis of $\zeta(3)/\pi^3$, numerical verification of integral representations, and targeted series identities mixing $\zeta(3)$ and π.

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6. Conclusion

No algebraic relation between $\zeta(3)$ and π was found within the tested bounds. Across 34 independent tests at precisions up to 20000 digits (extended to 40000 digits for the main result), every PSLQ computation returned a certified null result.

The two headline results: $\zeta(3) \neq (a/b)\cdot\pi^2 + c/d$ with coefficients up to 10^{18695}, and $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with height up to 10^{100}. These bounds far exceed any known identity in zeta function theory - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

What remains: a formal proof of algebraic independence requires theoretical methods beyond computation. But our results establish that if any relation exists, it lives in a regime (degree > 30, coefficients > 10^{18695}) that has no precedent in number theory. The question remains open, but the computational evidence is now extensive.

Any algebraic relation between $\zeta(3)$ and π, if it exists, must involve either coefficients larger than 10^{18695} or degree higher than 30.

References

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Appendix A: PSLQ Test Summary

All tests run on Apple M3 (8 cores). Times are for the PSLQ step only. Test numbers match Section 3.2.

Category 1: Linear independence from π

#	Basis	Size	Digits	Bound	Time	Result
1	**$\{\zeta(3), \pi^2, 1\}$**	**3**	**20000**	**10^{2000}**	**454s**	**No relation**
1'	$\{\zeta(3), \pi^2, 1\}$ (extended)	3	40000	10^{18695}	10484s	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	0.40s	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	0.24s	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	0.11s	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	0.83s	No relation

Category 2: Algebraicity of ζ(3)

#	Basis	Size	Digits	Bound	Time	Result
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	0.11s	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	0.17s	No relation

#	Basis	Size	Digits	Bound	Time	Result
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	0.25s	No relation

Category 3: Algebraicity of $\zeta(3)/\pi^3$ and bivariate tests

#	Basis	Size	Digits	Bound	Time	Result
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	12.6s	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	34.8s	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	6000	10^{200}	27.0s	No relation
12	**$\{(\zeta(3)/\pi^3)^k : k=0..30\}$**	**31**	**4500**	**10^{100}**	**29.4s**	**No relation**
13	$\{\zeta(3)^i\pi^j : i+j\leq3\}$	10	5000	10^{12}	4.8s	No relation
14	$\{\zeta(3)^i\pi^j : i+j\leq4\}$	15	5000	10^8	11.0s	No relation
15	$\{\zeta(3)^i\pi^j : i+j\leq6\}$	28	4000	10^{50}	18.1s	No relation

Category 4: Odd zeta values and other constants

#	Basis	Size	Digits	Bound	Time	Result
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	0.15s	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	0.35s	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	0.23s	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	0.89s	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	0.53s	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	0.48s	No relation
22	$\{\zeta(3), G^2, G\cdot\pi, \pi^2, G, 1\}$	6	3000	10^{10}	0.58s	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3)\cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	0.50s	No relation
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	0.59s	No relation
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	0.77s	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	0.50s	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	0.10s	No relation

Category 5: L-values and BBP-type tests

#	Basis	Size	Digits	Bound	Time	Result
28	$\{\zeta(3), L(E_{32},2), \pi^2, 1\}$	4	2000	10^{10}	0.25s	No relation

#	Basis	Size	Digits	Bound	Time	Result
29	$\{\zeta(3), L(\chi_{-4},3), \pi^2, 1\}$	4	2000	10^{10}	0.26s	No relation
30	$\{\zeta(3), L(E_{32},2), L(\chi_{-4},3), \pi^2, 1\}$	5	2000	10^8	0.35s	No relation
31	$\{\zeta(3), \pi^2\ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	0.89s	No relation
32	$\{\zeta(3), \mathrm{Li}_3(1/3), \pi^2\ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	0.30s	No relation
33	$\{\zeta(3), \mathrm{Li}_3(1/4), \pi^2\ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	0.31s	No relation

Category 6: Multiple zeta values

#	Basis	Size	Digits	Bound	Time	Result
34	$\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$	5	4000	10^{10}	0.74s	No relation

Category 7: Validation (known identities recovered)

#	Basis	Size	Digits	Bound	Time	Result
35	$\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$	4	5000	10^6	0.14s	**FOUND** [21,-24,-2,4]
36	$\{\zeta(3), \mathrm{Li}_3(-1)\}$	2	5000	10^6	<0.01s	**FOUND** [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	0.01s	**FOUND** [0,-945,1,0]

Appendix B: Continued Fraction Data

Continued fraction of $\delta = \pi^2/8 - \zeta(3)$, first 100 partial quotients:

```
[0; 31, 1, 1, 1, 1, 20, 4, 3, 1, 1, 11, 1, 4, 23, 9, 39, 1, 1, 6,
 1, 1, 35, 1, 7, 6, 1, 2, 2, 1, 1, 1, 1, 5, 1, 1, 11, 1, 2, 6, 1,
 5, 23, 2, 2, 2, 7, 2, 6, 155, 2, 1, 1, 59, 1, 3, 1, 16, 9, 1, 3,
 1, 1, 1, 4, 13, 5, 1, 4, 4, 4, 1, 3, 1, 3, 3, 1, 3, 1, 4, 3,
 4, 11, 2, 1, 1, 2, 18, 7, 1, 1, 15, 2, 1, 2, 71, 1, 4, 1, 1, 8]
```

Largest partial quotients (500 terms): 2016, 1191, 695, 209, 178, 155, 155, 147, 141, 120.

Appendix C: Computational Reproducibility

The complete set of tests can be reproduced by running `run_tests.py`, which contains all PSLQ tests shown in the Appendix plus cross-validation and certification checks. Every result reported in this paper is generated by that script.

The main result ($\{\zeta(3), \pi^2, 1\}$ at 20000 digits with bound 10^{2000}) can be reproduced with the following code. Note the explicit `maxsteps`: mpmath's default of 100 iterations is far too few to reach a norm bound anywhere near 10^{2000}.

```
from mpmath import mp, zeta, pi, pslq

mp.dps = 20000
z3 = zeta(3)
result = pslq([z3, pi**2, mp.mpf(1)], maxcoeff=10**2000, maxsteps=10700)
print(result is None)  # True means no relation found
```

This takes approximately 7.6 minutes on an Apple M3 processor. The extended verification (Result A, extended) uses the same code with `mp.dps = 40000` and `maxsteps = 100000`, taking approximately 2.9 hours. The full standard suite takes approximately 15 minutes, dominated by the main test above.

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